

LEBRE v0.52 — Specification and Technical Report

Online forecasting with statistically tested structural changes, under a minimal computational budget

LEBRE: Lifecycle-governed Evidence-Based Resource Evolution

Architecture: memory + structural expert + dynamic model averaging, with a plane of changes tested by e-processes

COST

~415 operations/step (analytic); 3.2–6.0 thousand instructions/step on a simulated Cortex-M4F

EVALUATION

three pre-registered evaluations on never-seen series (125, 40 and 60)

LAST EVALUATION (60 SERIES)

error $0.81 \times$ NLinear; tie with SARIMAX-X; ~25% above Chronos-2; 0 severe failures

STATUS

frozen research version, not promoted; scope: hydrology and buildings, one step ahead

1. Executive summary

LEBRE is an **online** forecaster for time series with inputs: at every step it receives the inputs and, when available, the target value; it predicts the next value, learns from the error and moves on, without re-processing the past and without prior training. It is designed to fit a computational budget of **a few hundred floating-point operations per step**, compatible with microcontrollers.

The central idea of this version is that **every structural change is an experiment**. The live model is simple (a memory of the target itself and a linear regression on the inputs). A challenger, running in shadow, proposes to add, remove, swap or refine the response to a **whole input**. The change is only accepted when an *e-process* — a sequential test valid at any time — accumulates enough evidence of predictive improvement, within a control of the rate of false changes over the stream. None of these parts is new in isolation; the document describes the combination, the conditions under which it works and what was measured.

Premise: **LEBRE does not aim to be the most accurate forecaster.** The goal is to deliver useful forecasts, without severe failures, at minimal per-step cost, without per-series tuning and with every structural change logged and justified. Expensive models are treated as **reference ceilings**, not as opponents.

What was measured (pre-registered evaluation on 60 never-seen series: 30 Brazilian river basins and 30 building meters):

- mean squared error relative to the online NLinear of **0.812** [IC 95% 0.752; 0.866] — the lowest among the eight comparators of similar cost evaluated in this protocol, and with no series with a severe failure;
- a tie with a SARIMAX with inputs fitted series by series (paired ratio 1.002); error ~25% **higher** than Chronos-2 with covariates (ratio 1.25), a foundation model ~10⁷ times more expensive;
- lower error than TTM (Tiny Time Mixers), zero-shot and fine-tuned with the inputs (ratios 0.747 and 0.771);
- mean analytic cost of 415 operations per step (maximum 1,020) — about 4% and 2% above the declared contract (400 and 1,000);
- on a **simulated** Cortex-M4F, in single precision: 3.2–6.0 thousand instructions per step on average (~31–57 μ s at 168 MHz, estimated) and 73 KB of state.

Main limits: results on two domains (hydrology and buildings) with few inputs, one-step forecasting; the testing mechanism **did not improve accuracy** compared to switching all inputs on without tests — its value lies in the logged and controlled structure; discovery fails on very smooth inputs with medium or long delays (information limit); the microcontroller cost was measured in a simulator, not on a board, and energy was not measured. The core of the mechanism (structural change accepted by an *e-process* of predictive improvement) already exists in the literature in another domain (§17).

Document guide

1. Executive summary	\$1	12. Development results	\$12
2. The problem, the premise and the scope	\$2	13. Evaluations on never-seen series	\$13
3. Design principles	\$3	14. Comparison with ultralight and foundation models	\$14
4. Architecture overview	\$4	15. Execution on a simulated microcontroller	\$15
5. Theoretical basis	\$5	16. Ablations: what each part contributes	\$16
6. Lifecycle of a structural change	\$6	17. Position in the literature	\$17
7. The condition of use: reference misadjustment	\$7	18. Evolution of the architecture	\$18
8. Output safeguards	\$8	19. Limits and claims we do not make	\$19
9. Analytic computational cost	\$9	20. Parameters	\$20
10. Observability and explainability	\$10	21. Pseudocode and reproducibility conditions	\$21
11. Evaluation methodology	\$11	22. References	\$22

2. The problem, the premise and the scope

The problem

Many systems continuously measure a target (river flow, building consumption, feeder load) together with inputs that influence it with delay (rain, temperature, upstream flows). Forecasting the next value *on the device itself* requires three things at once: very low per-step cost, operation without per-series tuning or prior training, and a behaviour that does not fail severely when the stream has gaps or changes. The most accurate modern models (foundation models) cost 10^7 to 10^{10} operations per forecast; classical models fitted series by series need maximum-likelihood optimisation; and cheap online linear models do not know, by themselves, which inputs and delays matter.

The premise

LEBRE occupies a specific point of the accuracy–cost–transparency trade-off. It is evaluated **within its budget class** (models with up to $\sim 10^4$ operations per forecast), with the expensive models as ceilings. “Delivering well” means: competitive error within the class, **no catastrophic failure**, predictable cost and an explanation of every change.

Scope of this version

- **one-step-ahead** forecasting, online, with the current inputs available (as in control and very-short-term forecasting);
- series with **few inputs** (1 to 5 in the evaluations) and an approximately linear relation with delays of up to 31 steps;
- the only information from the environment is the sampling period (e.g. 24 for hourly data with a daily cycle; for buildings, also 168 for the weekly cycle);
- inputs standardised causally (exponential mean and variance), as an input contract.

What LEBRE is not

It is not a foundation model nor a long-horizon forecaster; it does not model strongly non-linear relations; it does not claim physical causality (an accepted input is an input with certified *predictive relevance* against the reference, and near-identical inputs are reported as a group). The name comes from the project's original family of architectures (*Lifecycle-governed Evidence-Based Resource Evolution*): structure that enters and leaves the model by evidence, within a budget.

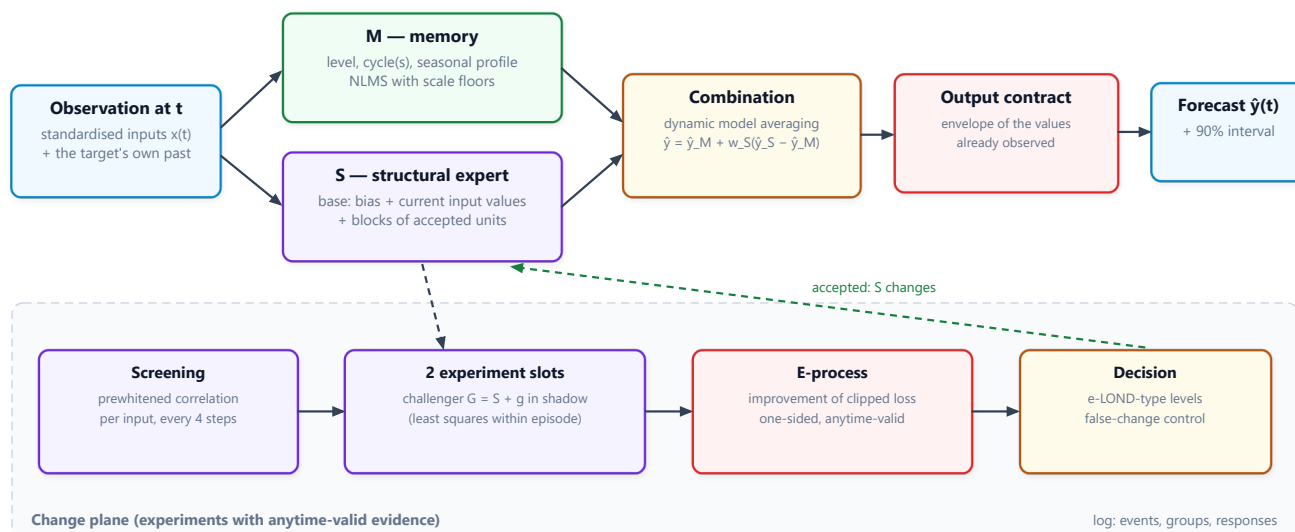
3. Design principles

Principle	Meaning
Every change is an experiment	Structure only changes when an anytime-valid sequential test rejects “the change does not improve the forecast”; each acceptance has an error level fixed beforehand.
Unit = a whole input	The question is “does this input help?”, not “does this lag help?”: fewer hypotheses, more power, the response being refined later by hierarchical splits.
Cost contract	Mean ≤ 400 and maximum $\leq 1,000$ floating-point operations per step on real series (revised during development, from 250 to 400, and recorded as such).
No per-series tuning	The same constants on any series; the environment only provides the period(s).
Silence and out-of-contract data are not evidence	Without an observed target, with a clipped input ($ x > 8\sigma$) or in quarantine, nothing learns and no evidence accrues.
Output within contract	The forecast never leaves an envelope of already observed values, and target gaps do not close the memory loop.
Observability	Every change is logged with time, kind, evidence and group of equivalent inputs; the forecast comes with an adaptive interval.

4. Architecture overview

D1 — LEBRE architecture: live forecast + shadow experiments

The forecast path (top) is cheap and always active; the change plane (bottom) proposes, tests and decides structural changes

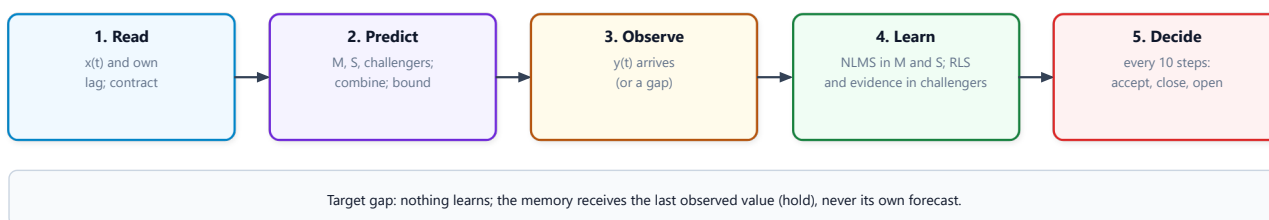


The two experts and the combination form the forecast path. The change plane runs in shadow and only alters S when a change is accepted.

Memory (M). Predicts the target from its own past, relative to the last value: reversion to a slow mean, the last difference, seasonal terms of the declared cycle(s) and a profile of increments per cycle phase. **Structural expert (S).** Linear regression on the current value of each input (the "base") plus the *response blocks* of the accepted units. **Combination.** Dynamic model averaging between M and S, with a weight that follows which of the two has been erring less. **Change plane.** A cheap screen points to the most promising candidate; up to two experiments run in shadow; an e-process accrues the evidence of each change; an online level procedure decides.

D2 — What happens in one time step

Strictly causal order: everything that decides the forecast at t uses only data up to $t-1$



Order of operations in one step. The forecast at t is issued before $y(t)$ is read.

5. Theoretical basis

This section gathers the theoretical basis used. **All tools come from the literature** (cited in each item); what is specific to LEBRE is the way they are combined and the verified conditions of use (§7).

5.1 Online linear learning: NLMS and recursive least squares

The live model (base of S, block weights and memory) learns by **NLMS** (normalised least mean squares; Widrow & Stearns 1985; Haykin 2002):

$$w_{t+1} = w_t + \mu \frac{e_t}{\|\phi_t\|^2 + \delta} \phi_t, \quad e_t = y_t - w_t^\top \phi_t.$$

In steady state, the **misadjustment** (relative excess squared error caused by the weight jitter) is approximately $M \approx \mu / (2 - \mu)$. LEBRE uses $\mu = 0.05$ in S and in the memory.

The **challengers** use recursive least squares (RLS) without forgetting within each episode, with $P_0 = 10 I$: with few parameters (5 to 7), RLS converges in tens of samples and gives an almost constant coefficient within the episode, which matters for the condition of use (§7).

5.2 Memory: exponential smoothing in regression form

The memory is a regression of the increment on features of its own past, inspired by the *airline* model (Box & Jenkins 1970) and by the Holt-Winters seasonal state (Hyndman et al. 2008) — an analogy, not an equivalence:

$$\begin{aligned} \hat{y}_t^M = & y_{t-1} + w_1(m_t - y_{t-1}) + w_2(y_{t-1} - y_{t-2}) + w_3(y_{t-s} - y_{t-1}) + w_4(y_{t-s+1} - y_{t-s}) + w_5 G_{t \bmod s} \\ & + [w_6(y_{t-s_2} - y_{t-1}) + w_7(y_{t-s_2+1} - y_{t-s_2})] \end{aligned}$$

m: slow exponential mean (rate 0.01); G: exponential mean (rate 0.1) of the increments observed at the same cycle phase; s_2 : second declared cycle (week, for buildings). The NLMS normalisation has scale-relative floors (10^{-3} of the mean feature power and 10^{-4} of the slow scale of y), so that flat stretches do not produce huge steps.

5.3 Combination: dynamic model averaging

The weight of S follows the discounted loss difference between the two experts (Raftery, Kárný & Ettler 2010; a logistic form equivalent to exponential expert weights):

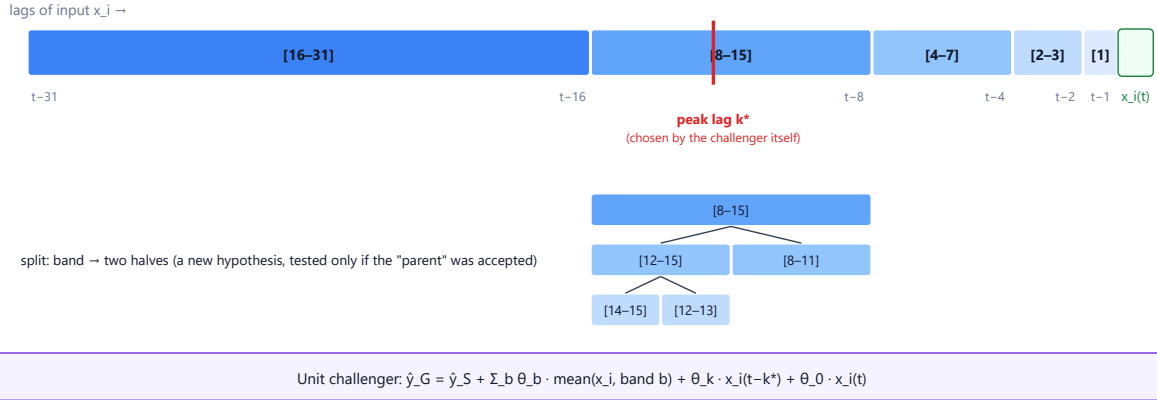
$$D_t = 0.99 D_{t-1} + \frac{(y_t - \hat{y}_t^S)^2 - (y_t - \hat{y}_t^M)^2}{\hat{\sigma}^2}, \quad w_S = \frac{1}{1 + e^{0.5 D_t}}, \quad \hat{y}_t = \hat{y}_t^M + w_S(\hat{y}_t^S - \hat{y}_t^M).$$

5.4 Response representation: octave bands, peak lag and Haar splits

The response of an input is approximated by means of its past over dyadic bands [1], [2–3], [4–7], [8–15], [16–31] — a resolution that decreases with delay, as in multiresolution decompositions (Mallat 1989) and distributed-lag models. A band mean dilutes an isolated delay (it keeps $\sim 1/\text{width}$ of the effect); therefore the challenger also carries its **peak lag** k^* , chosen by itself as the largest prewhitened correlation in the first 200 samples of the warm-up — before any evidence, which keeps the challenger predictable. After acceptance, each band may be **split** into its two halves (one Haar coefficient); each split is a new hypothesis, tested only if the “parent” was accepted, in the spirit of hierarchical testing (Meinshausen 2008; Yekutieli 2008).

D3 — Unit of change: a whole input, refined by splits

An input challenger uses means of the past over octave bands; hierarchical (Haar-like) splits refine the response after acceptance



Representation of the response to an input: means over octave bands, the current value and a peak lag; after acceptance, each band may be split into halves, one hypothesis per split.

5.5 Screening: prewhitening and group statistic

To choose the next candidate without testing them all, each inactive input has a group statistic: exponential correlations (weight 0.02) between the residual of S and the band means, both **prewhitened** by a causal AR(1) of the input (Box & Jenkins 1970), squared, summed and normalised by the null variance. The unit is opened if the statistic exceeds the 99% quantile of χ^2_5 (15.09); a split, the 99% quantile of χ^2_1 (6.63), on the Haar detail. The screen **produces no evidence**: it only chooses what to test, and the error control comes entirely from the e-process.

5.6 Anytime-valid evidence: e-processes of predictive improvement

An **e-process** is a non-negative process whose expectation under the null does not exceed 1 at any stopping time; by Ville's inequality, the probability that it ever exceeds $1/\alpha$ is $\leq \alpha$. This allows looking at the evidence at every step, pausing and resuming without inflating the error (Ramdas, Grünwald, Vovk & Shafer 2023; Grünwald, de Heide & Koolen 2024). LEBRE uses the sequential comparison of forecasters of Choe & Ramdas (2024), with the *weak* null: on average, the challenger does not improve the loss of the reference by more than a margin ϵ .

Clipped loss and differential, with the scale $B_t = 2 \max(\hat{\sigma}_t^S, 0.1 s_t^{\text{lenta}})$:

$$\ell_t(u) = \min\{(y_t - u)^2 / B_t^2, 1\}, \quad d_t = \ell_t(\hat{y}_t^S) - \ell_t(g_t) - \epsilon.$$

Sub-exponential mixture e-process, with predictable centring $\gamma_i = \min(\bar{d}_{i-1}, 0)$ and 8 values of λ :

$$E_n = \frac{1}{8} \sum_{j=0}^7 \exp \left\{ \lambda_j \sum_{i \leq n} d_i - \psi_{E,c}(\lambda_j) \sum_{i \leq n} (d_i - \gamma_i)^2 \right\}, \quad \psi_{E,c}(\lambda) = \frac{-\log(1 - c\lambda) - c\lambda}{c^2}, \quad \lambda_j = \frac{0.9 \cdot 2^{-j}}{c}.$$

The one-sided constant c . The statement of Choe & Ramdas assumes $|d_i - \gamma_i| \leq c$, which would give $c = 2(1+|\epsilon|)$. The underlying proof (Howard et al. 2021, §A.8) only uses the lemma of Fan, Grama & Liu (2015): $\exp\{\lambda \xi - \psi_E(\lambda) \xi^2\} \leq 1 + \lambda \xi$ for $\xi \geq -1$ and $\lambda \in [0, 1]$. For the one-sided test, under the weak null $E[d_i | \text{past}] = \mu_i \leq 0$, the **lower** bound $d_i - \gamma_i \geq -c$ suffices:

$$\mathbb{E}_{i-1} \left[e^{\lambda(d_i - \gamma_i) - \psi_{E,c}(\lambda)(d_i - \gamma_i)^2} \right] e^{\lambda(\gamma_i - \mu_i)} \leq (1 + \lambda(\mu_i - \gamma_i)) e^{\lambda(\gamma_i - \mu_i)} \leq 1.$$

Since $\lambda \Sigma d_i \leq \lambda \Sigma (d_i - \mu_i)$, the process is bounded by a non-negative supermartingale. Since $d_i \geq -(1+\epsilon)$ and $\gamma_i \leq 0$, $c = 1 + \max(\epsilon, 0)$. A mixture of e-processes is an e-process, and pauses between episodes form a predictable subsequence.

This observation is a direct reading of the published proof, not a new result. In a development simulation (400 trajectories $\times$ 5,000 steps, null at the boundary) the type-I error was ≤ 0.025 against $\alpha = 0.05$, and power rose from 0.66 (two-sided c) to 0.77 (Figure 5). The betting alternative with predictable λ (Waudby-Smith & Ramdas 2024) had more power but accepted near-identical substitutes of the true input and was not adopted.

5.7 Multiplicity over the stream: e-LOND-type levels

Each change hypothesis receives, at creation, a fixed level that depends only on the number of hypotheses already created and of changes already accepted (the LOND rule of Javanmard & Montanari 2018, in the e-value version of Xu & Ramdas 2024). This controls the false change rate (FDR) over an endless stream, with asynchronous decisions:

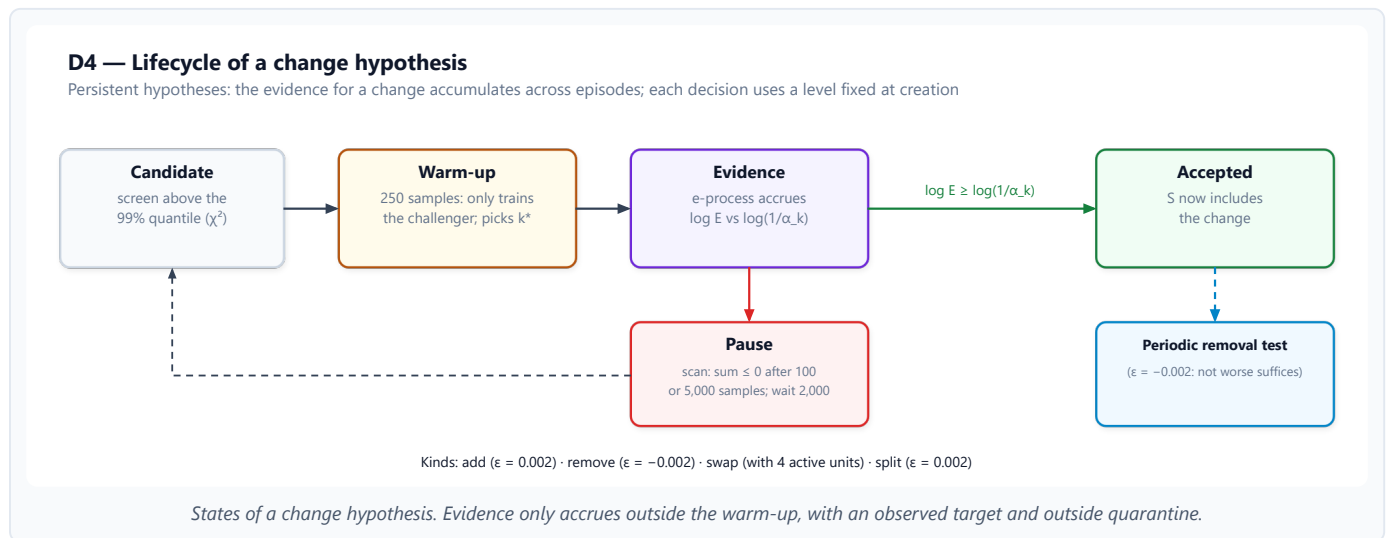
$$\alpha_n = \min(\alpha \gamma_n(R_n + 1), 0.5), \quad \gamma_n = \begin{cases} 1/(2P) & n \leq P \\ \frac{1}{2} C / ((n - P) \log^2(n - P + 1)) & n > P \end{cases}, \quad \text{accept if } \log E \geq \log(1/\alpha_n),$$

with $\alpha = 0.05$, $P = 2 \times (\text{number of units})$ and C such that $\Sigma \gamma_n \leq 1$. A hypothesis is *persistent*: its evidence accumulates across episodes.

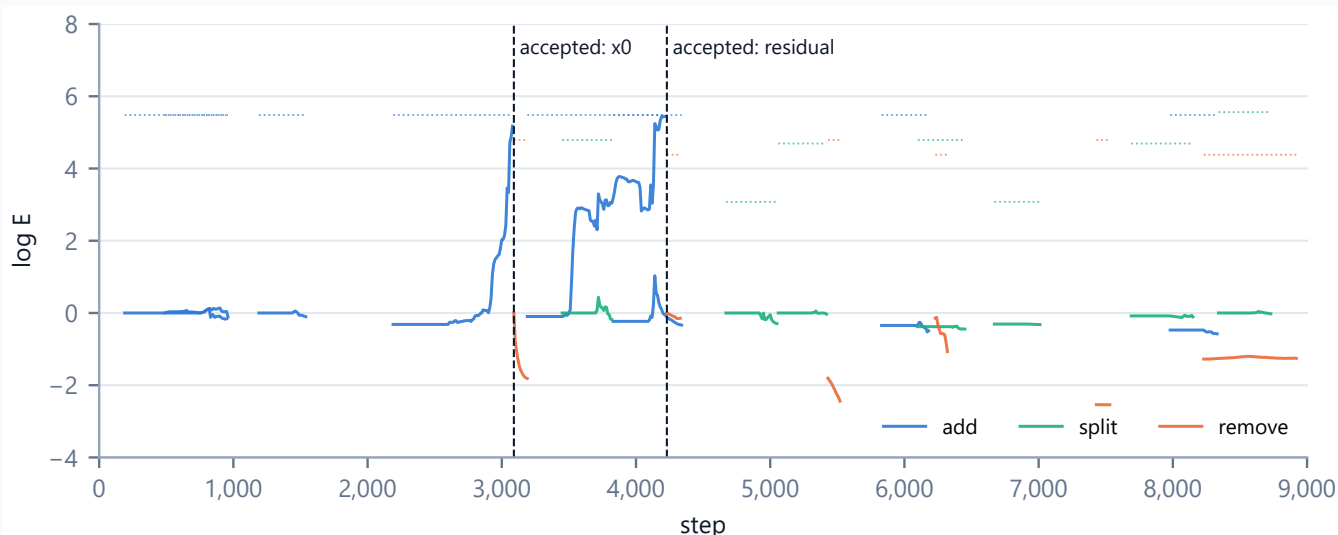
5.8 Adaptive interval

The 90% interval is a quantile of the absolute error updated by coverage feedback, in the spirit of adaptive conformal inference (Gibbs & Candès 2021): every 8 steps, $\hat{q} \leftarrow \max(0, \hat{q} + 0.1 \cdot \sigma \Sigma(\text{miss} - 0.1))$. There is no formal coverage guarantee under dependence; empirical coverage is reported.

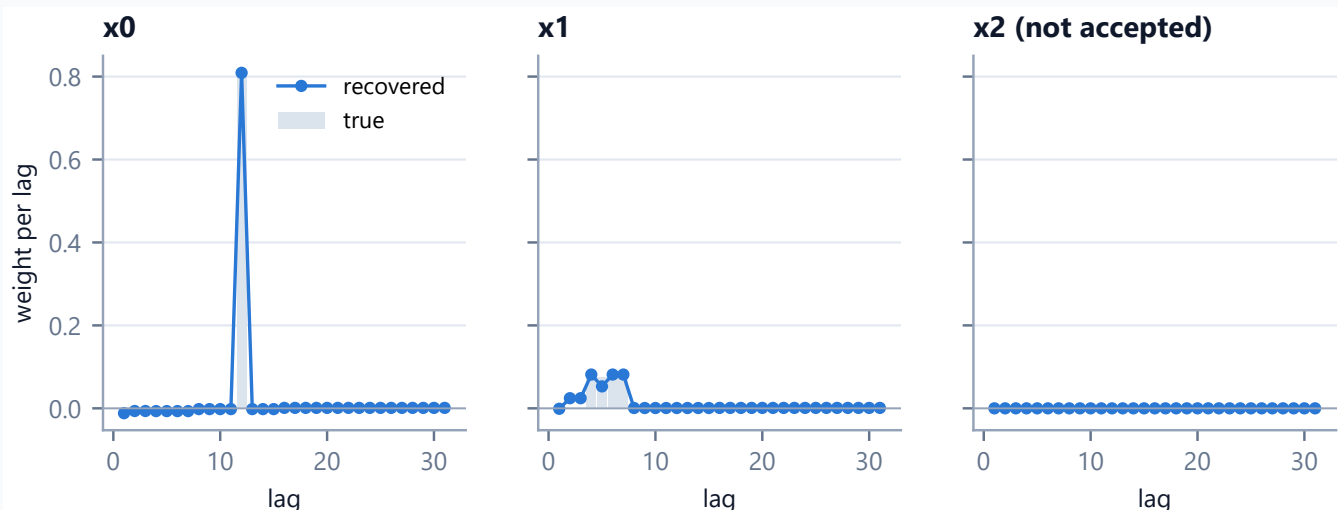
6. Lifecycle of a structural change



- **Units:** each input (including the target's own past, standardised causally) and the residual latent state (two filters of the standardised error, poles 0.8 and 0.95). Up to 4 active units.
- **Input challenger:** RLS on [means of the 5 bands, peak lag, current value $x_i(t)$] — the block is judged net of what $x_i(t)$ already carries. The RLS target is the residual the change would explain.
- **Episode:** 250-sample warm-up without evidence; ends on acceptance, by the scan rule (episode sum ≤ 0 after 100 samples) or at 5,000 samples; the same hypothesis waits 2,000 steps before returning. Only one challenger warms up at a time (cost-peak control).
- **Decisions** every 10 steps: accept, close, open. Priority: periodic removal test of active units; then the best screened candidate (unit or split). With 4 active units, the candidate enters by *swap* with the one of smallest contribution.
- **Judged on the forecast of S**, not on the combination: the question is whether the change improves the structural expert.



Real example (development series: flow at a hydro plant; input x_0 = outflow of the upstream plant; x_1 = the target's own past). Log-evidence of the experiments (solid) against their thresholds $\log(1/\alpha_k)$ (dotted). Input x_0 is accepted at $t \approx 3,090$ and the residual latent state at $t \approx 4,230$; afterwards, removals and splits are tested and closed without evidence.



Declared synthetic process ($y = 0.8 \cdot x_0(t-12) + 0.3 \cdot \text{mean}(x_1, t-4 \dots t-7) + \text{noise}$; x_2 irrelevant). Response recovered by LEBRE (dots) against the true one (bars): lag 12 of x_0 is recovered with weight 0.81; the distributed effect of x_1 appears in band [4–7]; x_2 is not accepted.

7. The condition of use: reference misadjustment

A test of predictive improvement compares **methods** (model + estimation procedure), not models — a distinction known in econometrics (Giacomini & White 2006). During development, this appeared concretely: with the base of S learning by NLMS with step 0.1, targets **without any structure** had a change accepted in ~20% of the runs. The cause is that the estimation error of the reference is predictable from the past, and a challenger can “correct” it.

Let $f_t^* = \mathbb{E}[y_t | \mathcal{F}_{t-1}]$, $\eta_t = f_t^* - S_t$ (estimation error of the reference, predictable) and the challenger $G_t = S_t + g_t$. For the normalised squared-loss difference $\Delta_t = [(y_t - S_t)^2 - (y_t - G_t)^2] / B_t^2$:

$$\textbf{Proposition 1} \quad \mathbb{E}[\Delta_t | \mathcal{F}_{t-1}] = \frac{2g_t\eta_t - g_t^2}{B_t^2} \leq \frac{\eta_t^2}{B_t^2}, \quad \text{with equality at } g_t = \eta_t.$$

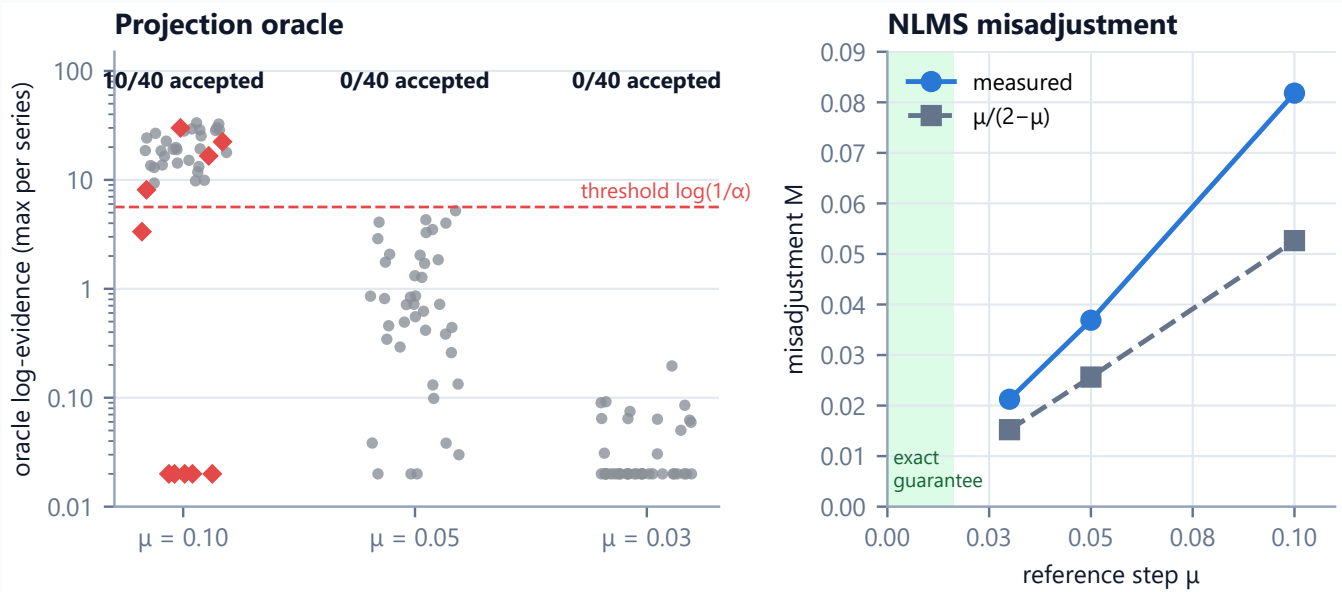
Proof: $y - S = \varepsilon + \eta$ and $y - G = \varepsilon + \eta - g$; expanding, the difference is $2g(\varepsilon + \eta) - g^2$, and $\mathbb{E}[\varepsilon | \text{past}] = 0$. ■

Corollary (exact guarantee) $\frac{1}{n} \sum_{i \leq n} \frac{\eta_i^2}{B_i^2} \leq \varepsilon \quad \forall n \Rightarrow$ under the structural null, the test null holds.

For NLMS with misadjustment M , this requires $M \lesssim 4\varepsilon$ (with $\varepsilon = 0.002$: $\mu \lesssim 0.016$).

Proposition 2 (fixed coefficient) mean improvement over the window $\leq D_u + \varepsilon$, $D_u = \frac{1}{|W|} \sum_{i \in W} (P_u \eta_i)^2 / B_i^2 - \varepsilon$,

where P_u is the projection of η on the unit's features. **Proposition 3 (practical criterion, approximation):** log-evidence grows as $n \cdot D^2 / (2\sigma_d^2)$; on a series of length T , false acceptances are unlikely if $D_u < \sigma_d \sqrt{2 \log(1/\alpha_k) / T}$.



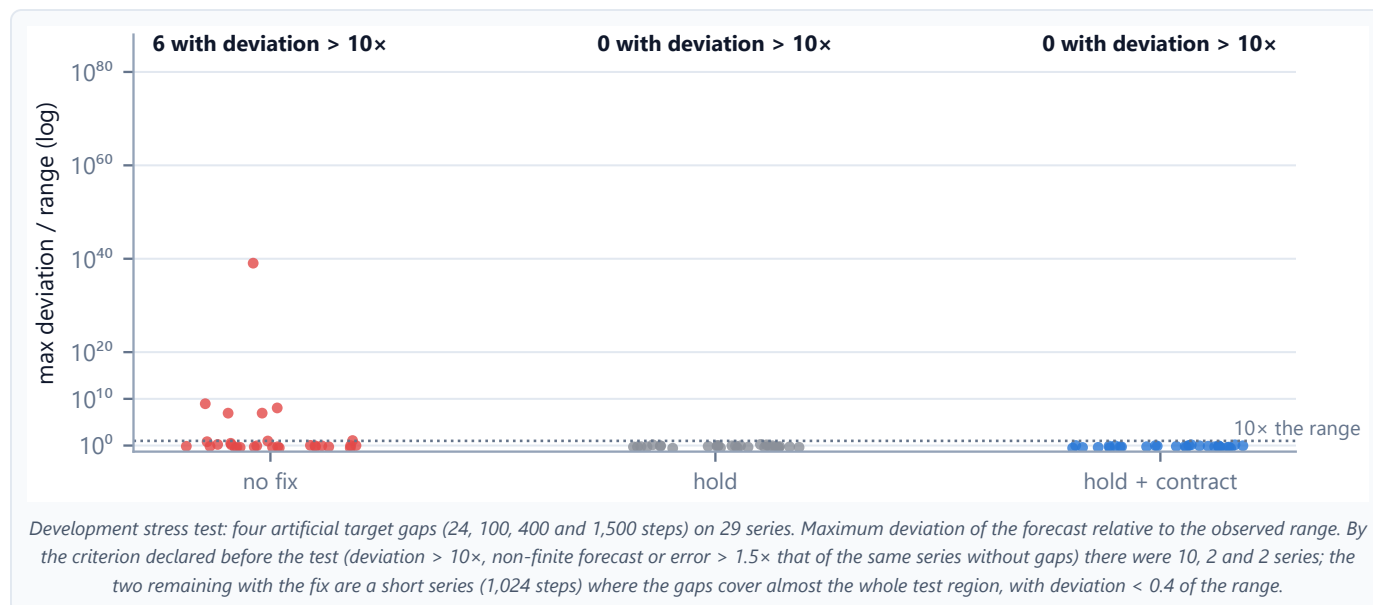
Development check (120 runs with structureless targets, real inputs). Left: log-evidence of the best possible challenger (projection oracle) against the threshold; actual acceptances in red. With $\mu = 0.1$ the oracle crosses the threshold (median 18.2) and there were 10/40 acceptances; with $\mu = 0.05$ (max 5.2) and $\mu = 0.03$, none. In runs that accepted, the oracle was measured after the acceptance and may be low (points at the floor). Right: measured and theoretical misadjustment.

Honest reading: LEBRE uses $\mu = 0.05$, which lies in the **practical regime** of Proposition 3 (the deviation from the null exists, ~ 0.0013 , but is smaller than what is detectable within the series length, ~ 0.0019), and **not** in the exact-guarantee regime ($\mu \lesssim 0.016$). Therefore false-change control is checked empirically with null batches (0/40 and 0/30 acceptances in development), not only asserted by theory. The observation relates to Giacomini & White (2006) and, through another mechanism, to Han & Qu (2026); we did not find it stated explicitly for this situation, but it is a simple consequence of known results.

8. Output safeguards

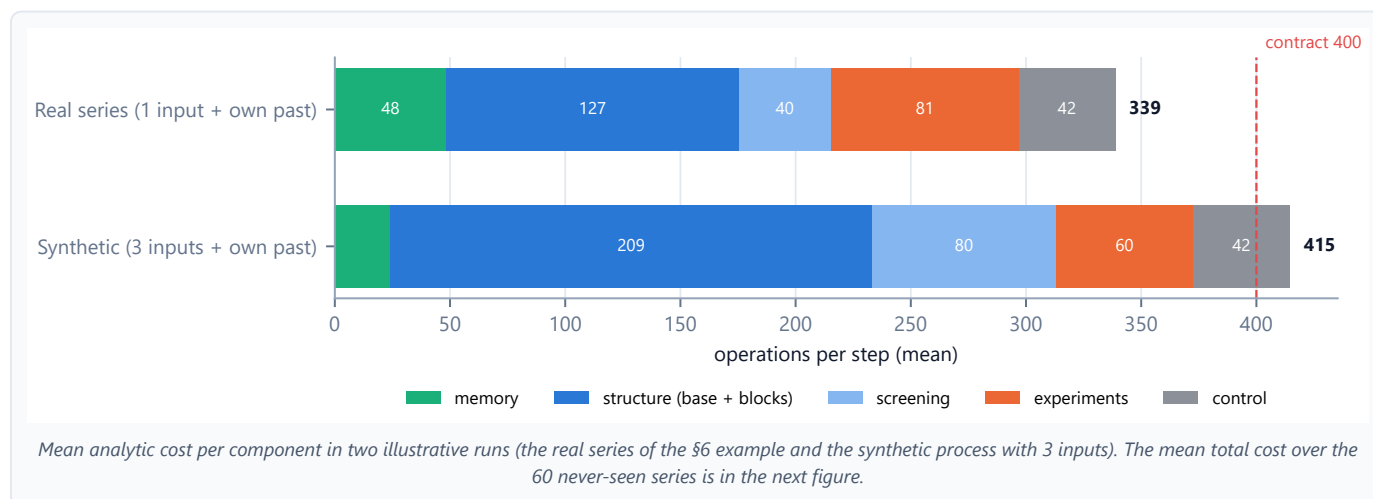
Two safeguards act only on the output path and cost ~ 15 operations per step. Both are standard engineering practice; they are here because their absence produced the most severe failure observed (§13).

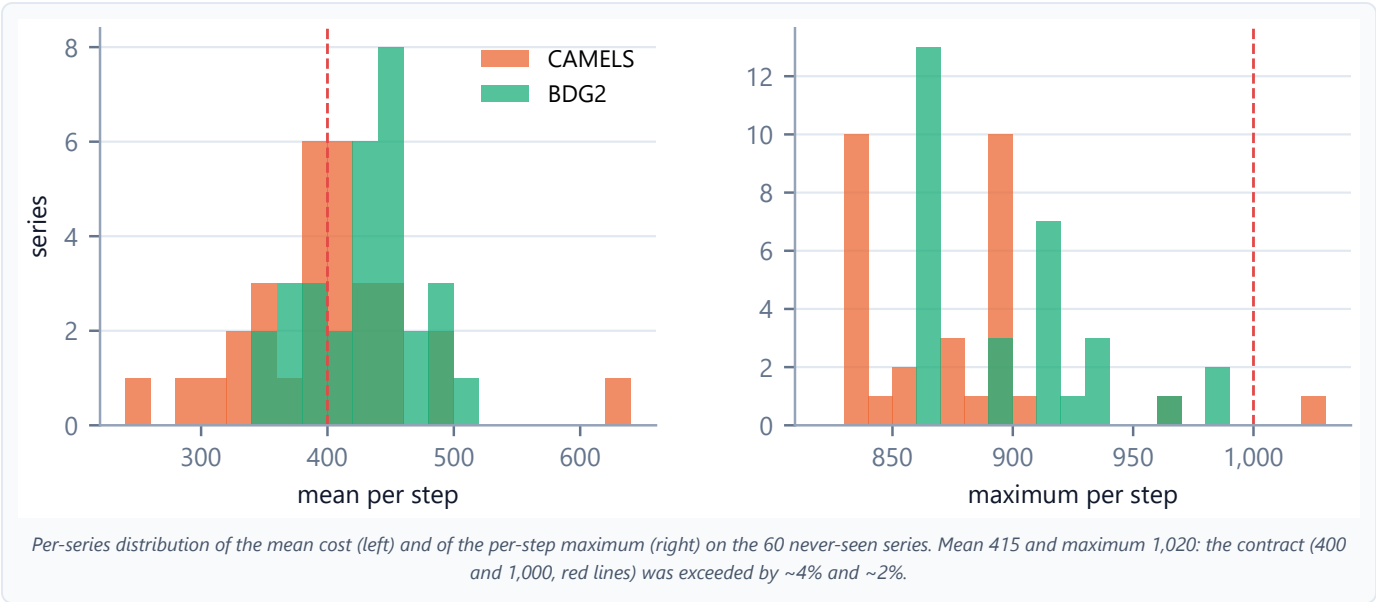
- **Hold during target gaps:** the memory receives the last observed value, never its own forecast. Without this, the memory runs open-loop and, with some learned weights, the recursion is unstable (characteristic root ~ 1.75 in a real case).
- **Output contract:** the forecast is limited to the envelope of observed targets (running maximum and minimum relaxing toward the slow mean at rate 10^{-3}), widened by $0.5\times$ its width; a non-finite forecast becomes the last observed value.



9. Analytic computational cost

Cost is counted operation by operation in the code (additions, products, divisions, roots, exponentials count 1), per component. The per-step cost is **almost fixed**: memory, base and screening always run; experiments add cost only when a slot is busy, and peaks occur at decision steps.





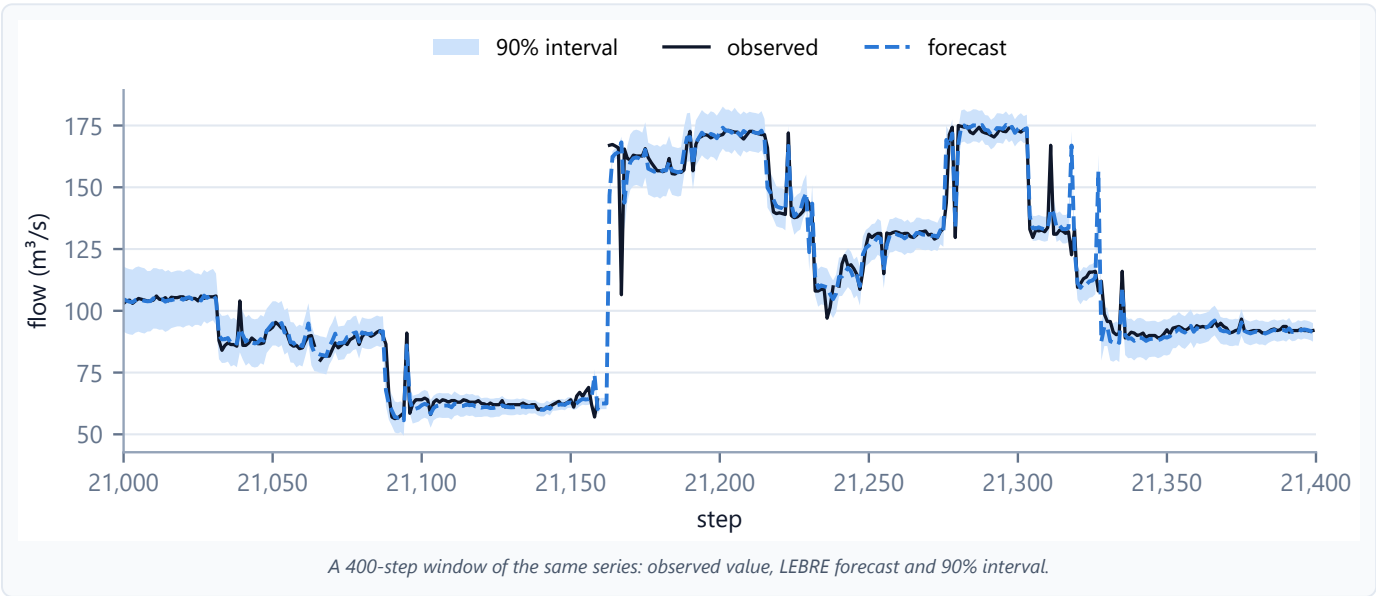
State memory has a fixed size, given the number of inputs and the periods. In the C port, with capacities sized for up to 8 inputs and 160 hypotheses, the state takes ~73 KB; most of it is the hypothesis table, which can be reduced (\$15).

10. Observability and explainability

Each forecast can be decomposed into named contributions (memory, base, each active block), and each structural change is logged. Example from the \$6 development series:

Time	Event	Change	log E	Evidence samples
3,090	accepted	add input x0	7.18	957
4,230	accepted	add residual latent state	5.49	893

- Equivalent groups:** together with the accepted input x0 the model reports the group {x0, x1} (causal correlation ≥ 0.95 between x0, the outflow of the upstream plant, and x1, the target's own past). The claim is "this input or an equivalent one": here, the upstream flow and the target's past carry almost the same information.
- 90% interval:** empirical coverage of 0.906 on this series and 0.904 on average over the 60 never-seen series.
- Log of closed experiments:** each test that did not pass is logged with its evidence (for instance, removal tests of x0 with log E between -1.8 and -3.2: the input remains useful).



11. Evaluation methodology

Protocol

- **Online one-step forecasting**, without looking ahead. All models receive **the same inputs**, standardised causally.
- **Metric**: mean squared error on the test region (last 70% of each series, observed target), divided by that of the calibrated online NLinear on the same series; geometric mean across series; 95% CI by bootstrap stratified by domain (2,000 resamples). Values below 1 = better than NLinear.
- **Catastrophic failure**: relative error > 10, deviation > 10× the observed range, or a non-finite forecast.
- **Foundation models** (too expensive to run at every step): evaluated on 1,000 points per series, the same for all, with the same past context and the current inputs as known future covariates.
- **Batch-fitted comparators** (SARIMAX with inputs, FITS, SparseTSF, TTM fine-tuning): fitted on the first 15% of the series (up to 5,000 points) and then used with fixed weights.

Data and split

Three public sources: flows and generation from the Brazilian National Grid Operator (ONS; cascaded plants, inputs = upstream flows), CAMELS-BR basins (Chagas et al. 2020; inputs = precipitation, evapotranspiration, temperature) and Building Data Genome 2 meters (Miller et al. 2020; inputs = weather), plus two system-identification benchmarks in development. Series were split by **fixed-seed permutations**, before any modelling, into development and reserves; the data loader blocks access to the reserves outside a pre-registered run.

Set	Series	Use
Development	29 (ONS 11, CAMELS 10, BDG2 5, Silverbox, Cascaded Tanks ×2)	all design and tuning; semi-synthetic and synthetic with declared seeds
Reserve 1	125 (ONS 45, CAMELS 50, BDG2 30)	pre-registered; revealed the gap failure (\$13.1)
Reserve 2	40 (CAMELS 20, BDG2 20)	pre-registered; checked the fix
Reserve 3	60 (CAMELS 30, BDG2 30)	pre-registered; ultralight, TTM, C port, microcontroller

Comparator classes

Class	Models	Cost per forecast
Budget	online NLinear and DLinear (Zeng et al. 2023), online Holt-Winters, dense ARX (NLMS), online LASSO, FITS (Xu et al. 2024), SparseTSF (Lin et al. 2024), LEBRE v0.51	20 to ~10 ⁴ operations
Classical reference	SARIMAX with inputs (maximum likelihood, series by series); least-squares ARX with order chosen online	optimisation-based fit; ~4.9·10 ⁴
Foundation models	TTM r2 (~1M parameters; Ekambaram et al. 2024) zero-shot and fine-tuned with inputs; Chronos-2 with covariates (Ansari et al. 2025)	15.6·10 ⁶ / 55.6·10 ⁶ ; ~10 ⁹ –10 ¹⁰ (estimated)

Pre-registration: before each access to a reserve, a document fixed questions, metrics, comparators, criteria and the code hashes; each reserve was run a single time. Infrastructure incidents (without any model change) are described with the results.

12. Development results

Results used to design the architecture. **They are not independent evidence**: they show the behaviour of the mechanisms under controlled conditions. Each iteration had a measurement set declared before running, with new seeds.



Structure discovery on controlled processes

Situation	Result (development)
Synthetic with white inputs (delays up to 12)	structure recovered to the exact delay in 500–2,200 steps (e.g. delay 12, weight 0.84 vs 0.80)
Semi-synthetic (real inputs, known effect)	true input (or its group) in 2 of 3 structured tasks, 3/3 seeds; latency 4–17 thousand steps
Very smooth inputs (autocorrelation ~0.99), medium/long delay	not discovered ; an online oracle (knows the delay, tests from the start) needs 9–27 thousand steps or never gets there in 32 thousand
Structureless targets (null batches with real inputs)	0/40 and 0/30 acceptances; 0 false exogenous inputs in all declared sets
Near-identical inputs (correlation 0.98)	indistinguishable by predictive improvement; the model reports the group

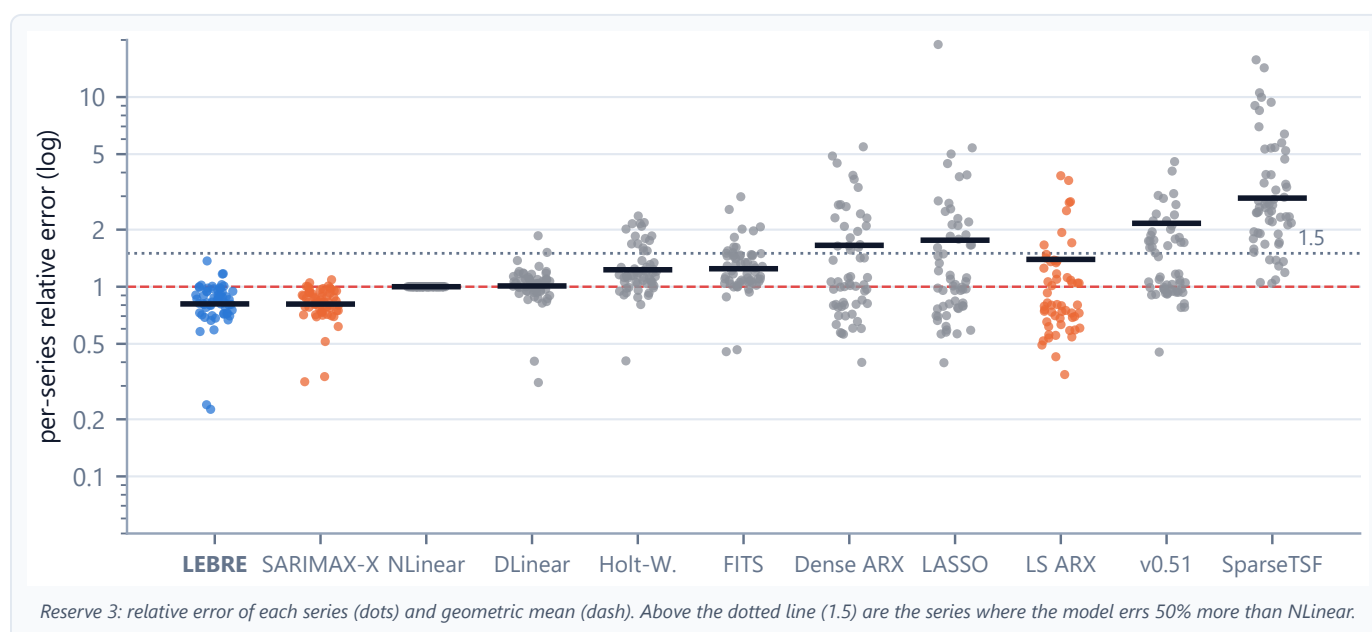
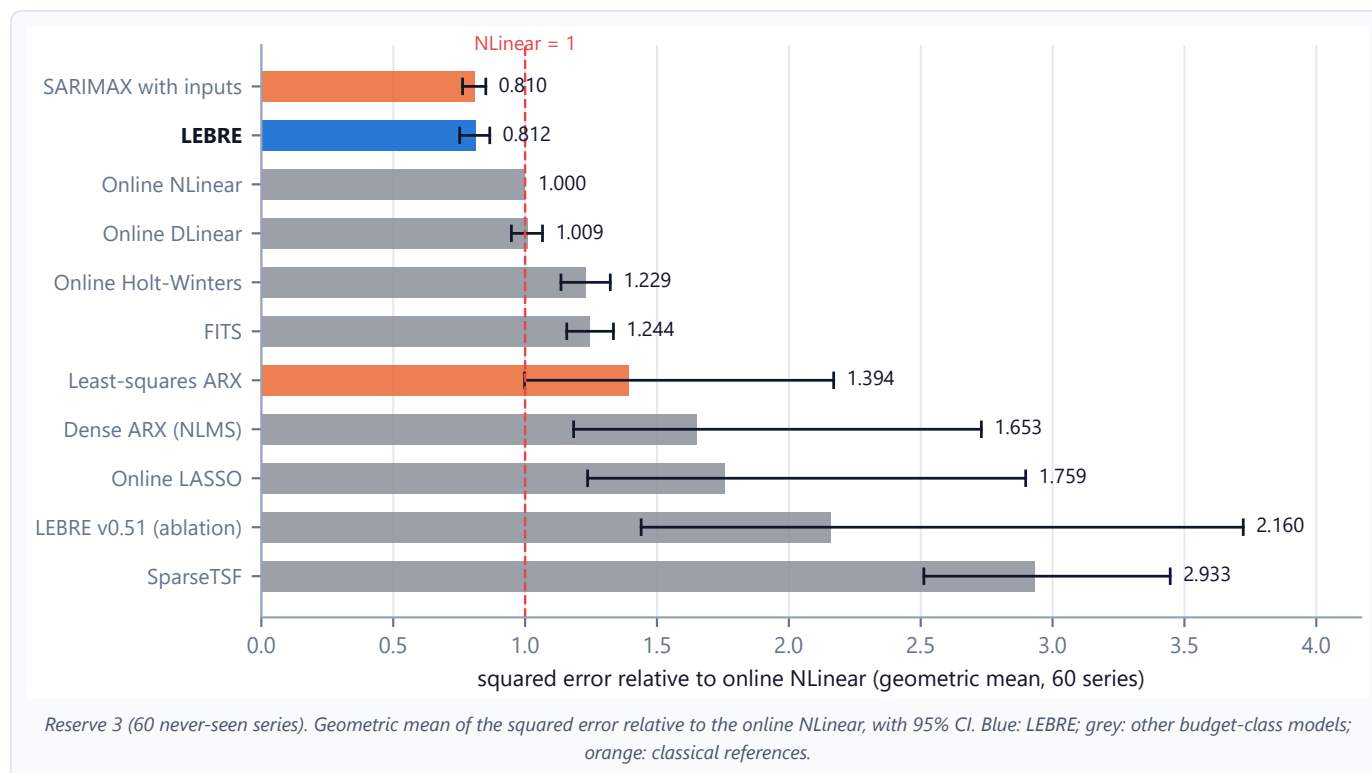
The limit on smooth inputs is known in principle: with very persistent regressors, the new information of a delay beyond the current value is small (Stambaugh 1999; Campbell & Yogo 2006). In those cases the forecast remains good because the base already uses the current input values; what is lost is the *explanation* of the delay.

13. Evaluations on never-seen series

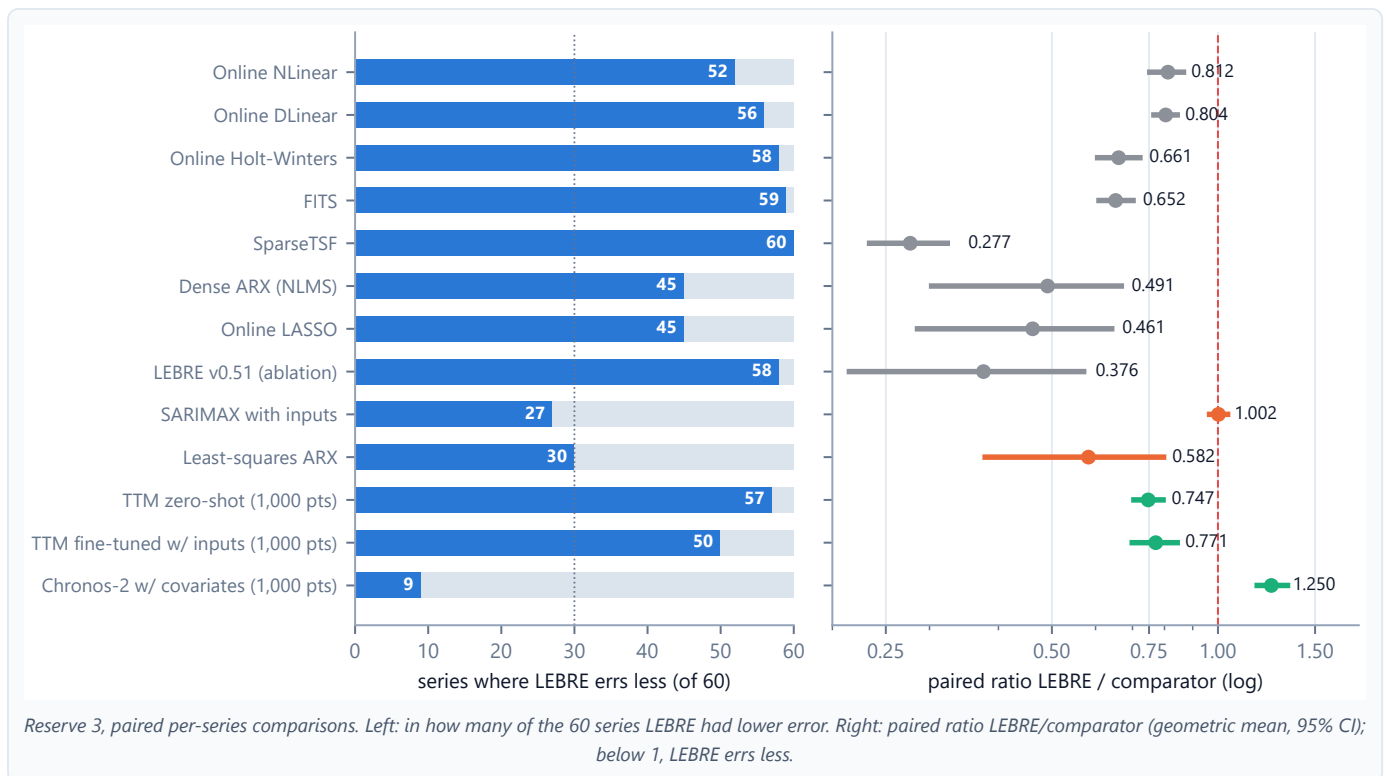
13.1 Reserve 1 (125 series): an engineering failure

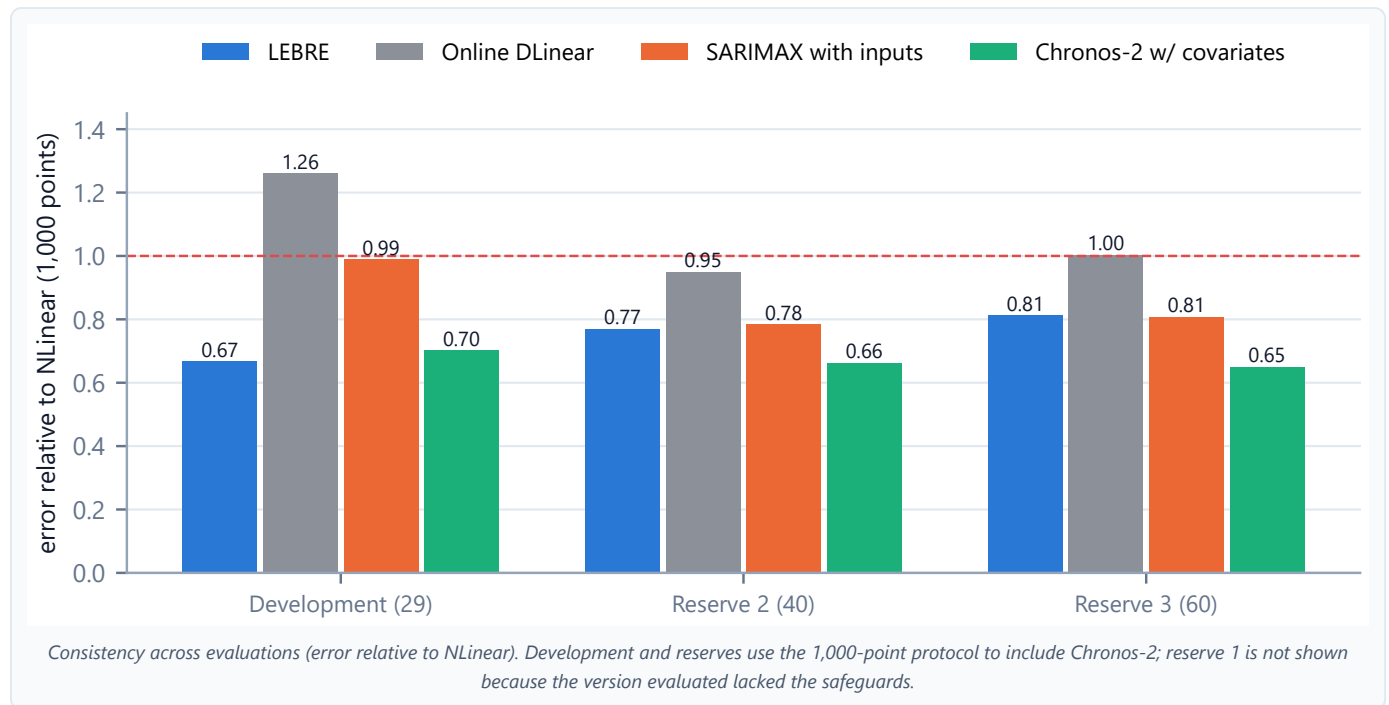
The first pre-registered evaluation, with the version still without the output safeguards, revealed that **6 of 125 series exploded** (relative error from 13 up to 10^{79}): when the target was missing for many steps, the memory filled its own history with its own forecasts. The pre-registered result was dominated by these series (geometric mean 5.20). In an exploratory view, *not pre-registered*, without the 6 series, the relative error was 0.706, equal to development. The failure motivated the safeguards of §8, which were checked in development and then on new never-seen series.

13.2 Reserves 2 and 3 (40 and 60 series)



Model	CAMELS	BDG2	All [95% CI]	Series > 1.5	Catastrophic	Operations/forecast
LEBRE	0.858	0.768	0.812 [0.752, 0.866]	0%	0	415
Online NLinear	1.000	1.000	1.000 [1.000, 1.000]	0%	0	674
Online DLinear	1.076	0.947	1.009 [0.948, 1.066]	3%	0	1,533
Online Holt-Winters	1.213	1.244	1.229 [1.136, 1.323]	25%	0	20
FITS	1.362	1.137	1.244 [1.158, 1.335]	18%	0	9,757
Dense ARX (NLMS)	1.008	2.711	1.653 [1.184, 2.730]	38%	2	474
Online LASSO	1.015	3.048	1.759 [1.237, 2.898]	38%	4	705
LEBRE v0.51 (ablation)	2.180	2.140	2.160 [1.440, 3.723]	48%	4	108
SparseTSF	2.980	2.886	2.933 [2.512, 3.446]	87%	3	4,016
SARIMAX with inputs	0.817	0.803	0.810 [0.763, 0.851]	0%	0	ML fit
Least-squares ARX	0.677	2.867	1.394 [0.998, 2.170]	20%	4	48,944

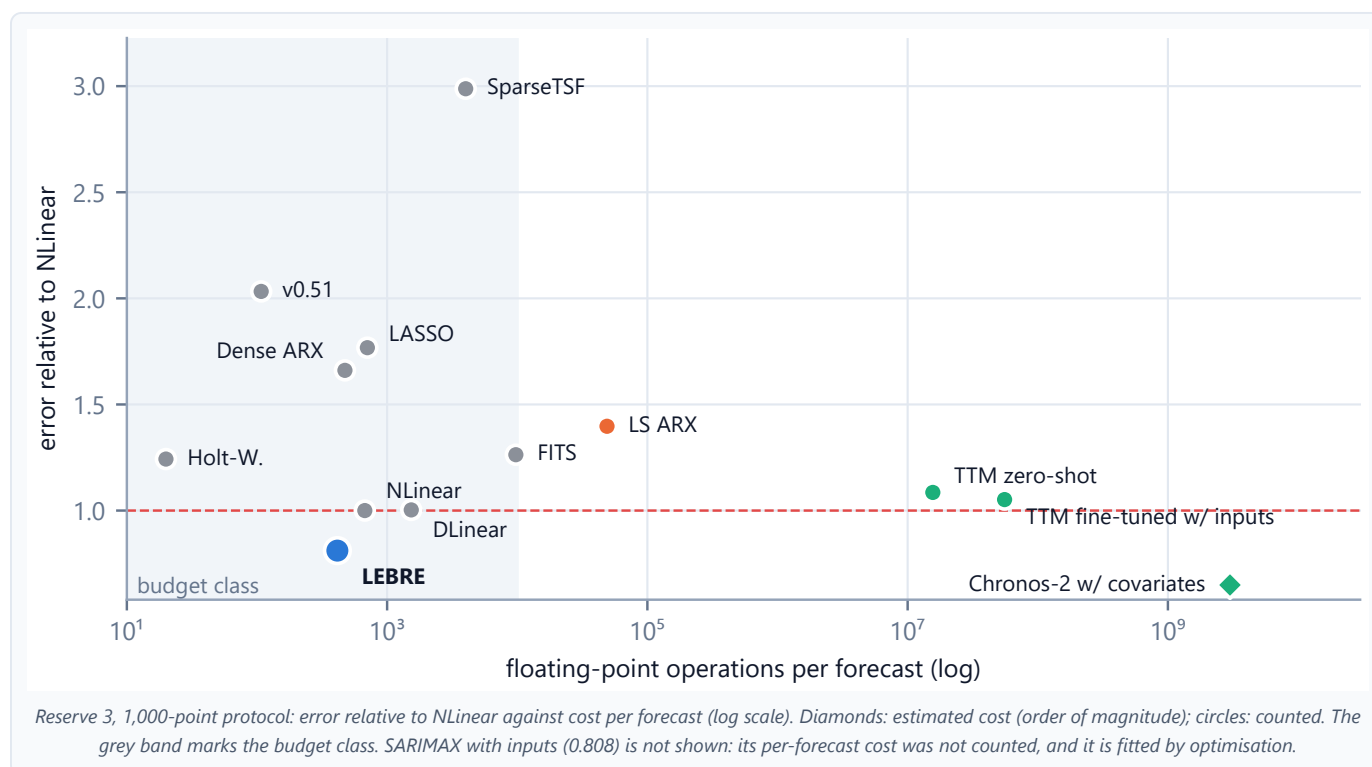




Summary of the two evaluations of the final version: relative error 0.767 (reserve 2) and 0.812 (reserve 3); no catastrophic series and none above 1.5 among the 100 series; a tie with SARIMAX with inputs in both (paired ratios 0.999 and 1.002); error 16% and 25% above Chronos-2 with covariates. The 90% interval had empirical coverage of 0.905 and 0.904. The table below summarises reserve 2 (it includes the version without the safeguards as an ablation).

Reserve 2 (40 series)	CAMELS	BDG2	All [95% CI]	Series > 1.5	Catastrophic	Operations/forecast
LEBRE	0.703	0.836	0.767 [0.657, 0.858]	0%	0	407
LEBRE without safeguards (ablation)	1.468	0.834	1.107 [0.746, 1.826]	8%	2	392
Online NLinear	1.000	1.000	1.000 [1.000, 1.000]	0%	0	674
Online DLinear	0.948	0.965	0.957 [0.812, 1.077]	5%	0	1,533
Online Holt-Winters	1.087	1.139	1.112 [0.946, 1.275]	18%	0	20
Dense ARX (NLMS)	0.916	4.345	1.995 [1.214, 3.836]	40%	4	474
Online LASSO	0.888	4.576	2.016 [1.235, 3.813]	42%	3	705
LEBRE v0.51 (ablation)	1.258	5.645	2.665 [1.161, 8.462]	32%	3	110
SARIMAX with inputs	0.703	0.837	0.767 [0.646, 0.870]	0%	0	ML fit
Least-squares ARX	0.595	3.387	1.419 [0.905, 2.597]	22%	3	48,944

14. Comparison with ultralight and foundation models

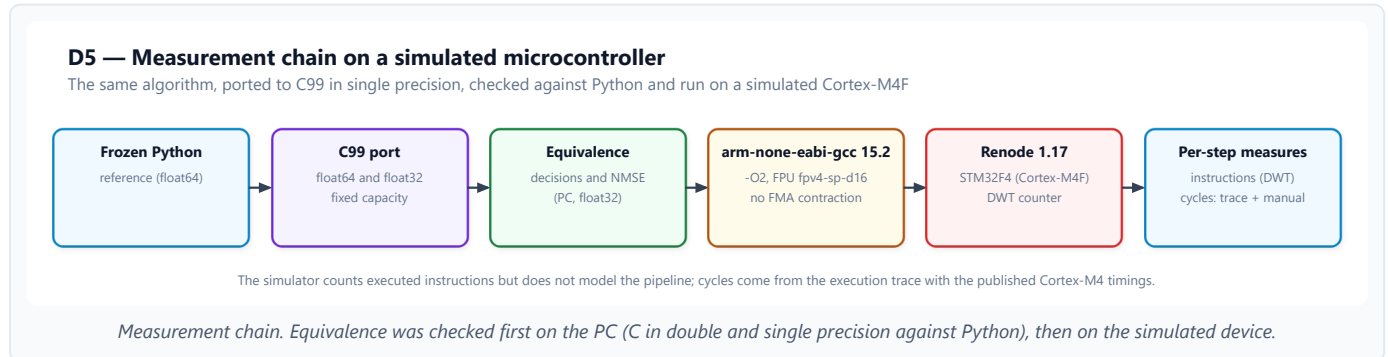


Model	CAMELS	BDG2	All	Paired ratio LEBRE/model [95% CI]	LEBRE wins	Operations/forecast
LEBRE	0.836	0.787	0.811	—	—	~415
SARIMAX with inputs	0.798	0.818	0.808	1.004 [0.956; 1.056]	28/60	ML fit
TTM zero-shot	1.204	0.978	1.085	0.747 [0.704; 0.795]	57/60	15.6·10 ⁶
TTM fine-tuned w/ inputs	1.049	1.054	1.052	0.771 [0.699; 0.844]	50/60	55.6·10 ⁶
Chronos-2 w/ covariates	0.782	0.539	0.649	1.250 [1.178; 1.337]	9/60	~10 ⁹ –10 ¹⁰

- **Ultralight (FITS, SparseTSF):** in the online one-step protocol, with fixed weights fitted on the first 15%, they were behind the online NLinear (1.244 and 2.933). This is not the regime they were designed for (long-horizon forecasting, batch training, univariate); the result only shows that, in *this* regime, parameter lightness does not translate into accuracy.
- **TTM:** two fine-tuning configurations were tried in development; the chosen one (rate 10⁻³, 5 epochs) was used in the reserve. Other fine-tuning setups may give different results.
- **Chronos-2 with covariates** is the reference ceiling: error ~25% lower than LEBRE overall, with the largest gap on buildings (0.539 vs 0.787).

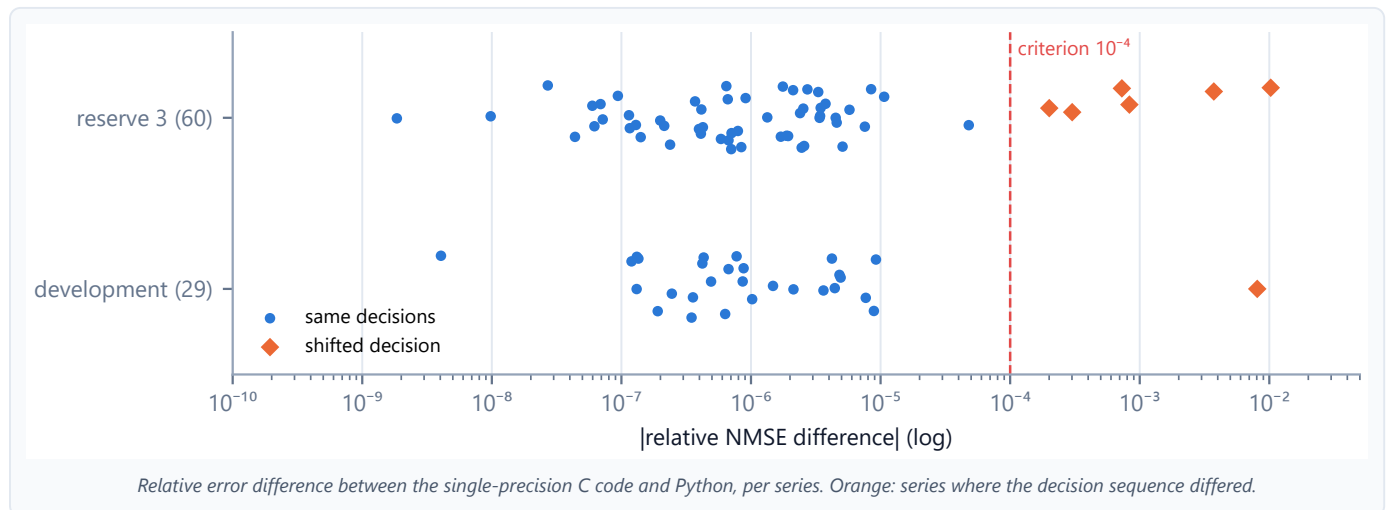
15. Execution on a simulated microcontroller

To check whether the analytic cost corresponds to something executable on a microcontroller, the frozen configuration was ported to C99 and run on a **simulator** of a common microcontroller (STM32F4, Cortex-M4F core at 168 MHz). There was no measurement on a physical board.



15.1 Port equivalence

In double precision, the C code exactly reproduces the Python decisions on 29/29 development series (relative error difference $\sim 10^{-16}$). In **single precision** (the microcontroller build), the sequences of accepted changes are identical on 28/29 development series and on 54/60 never-seen series; on the others, a decision near the threshold happens a few steps earlier or later (rounding). The median error difference was $\sim 10^{-6}$.

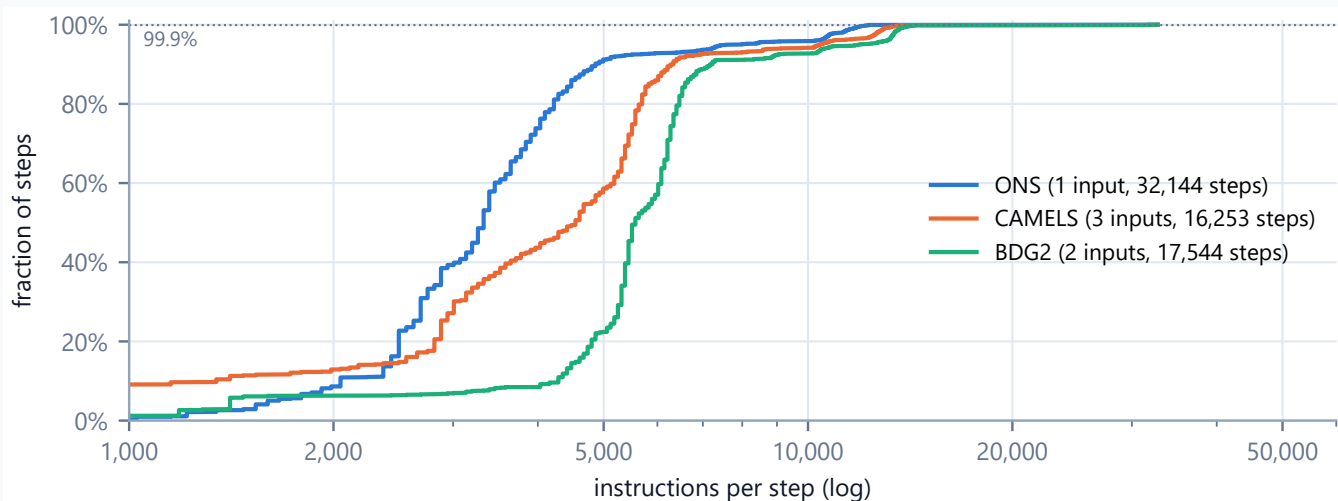
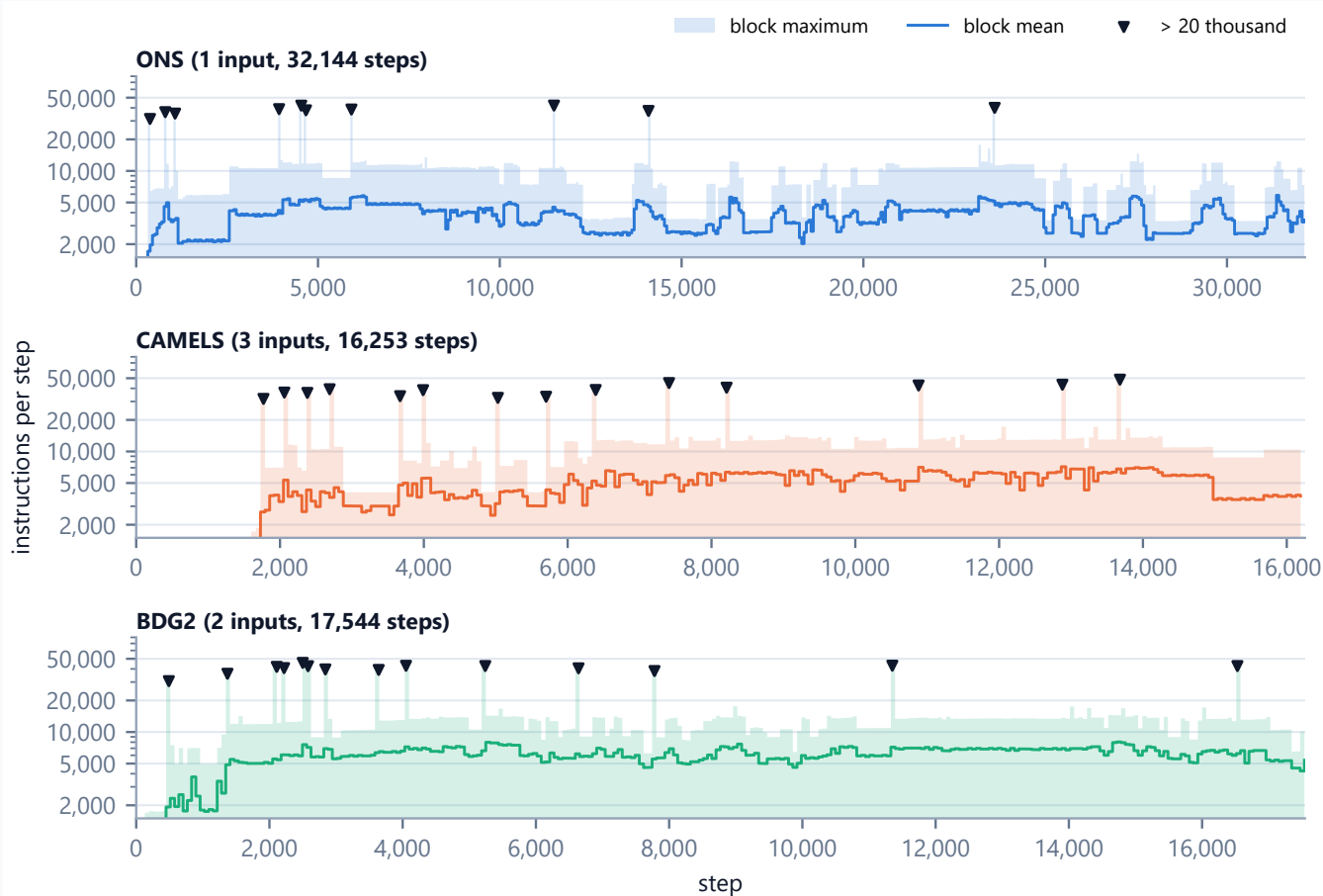


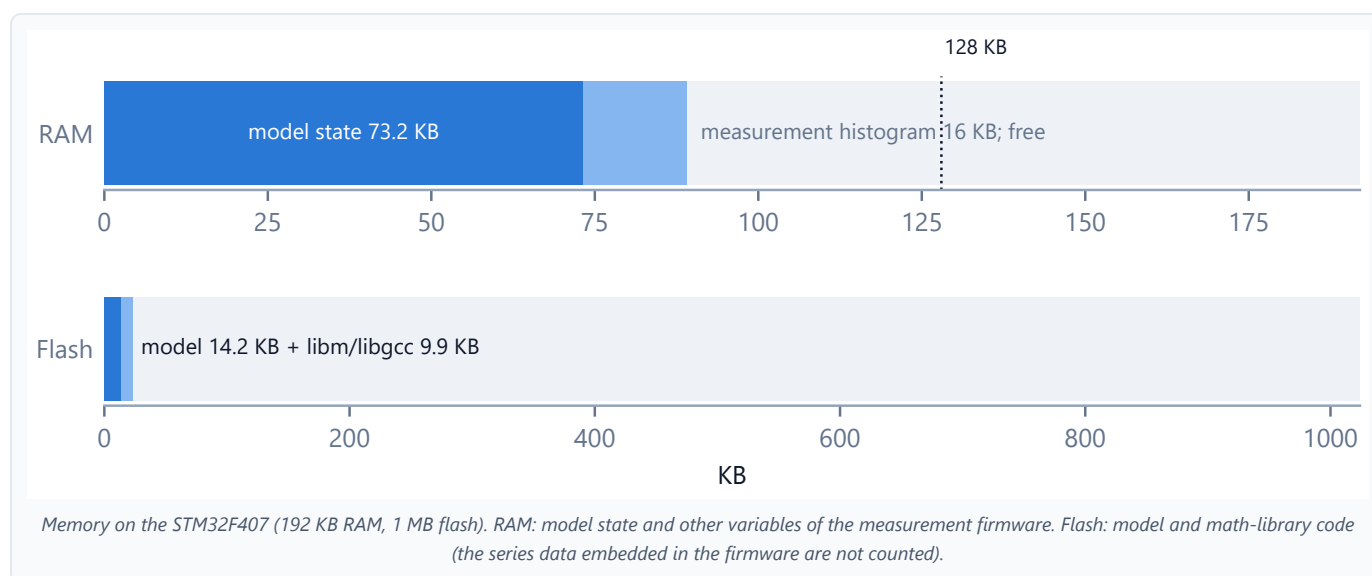
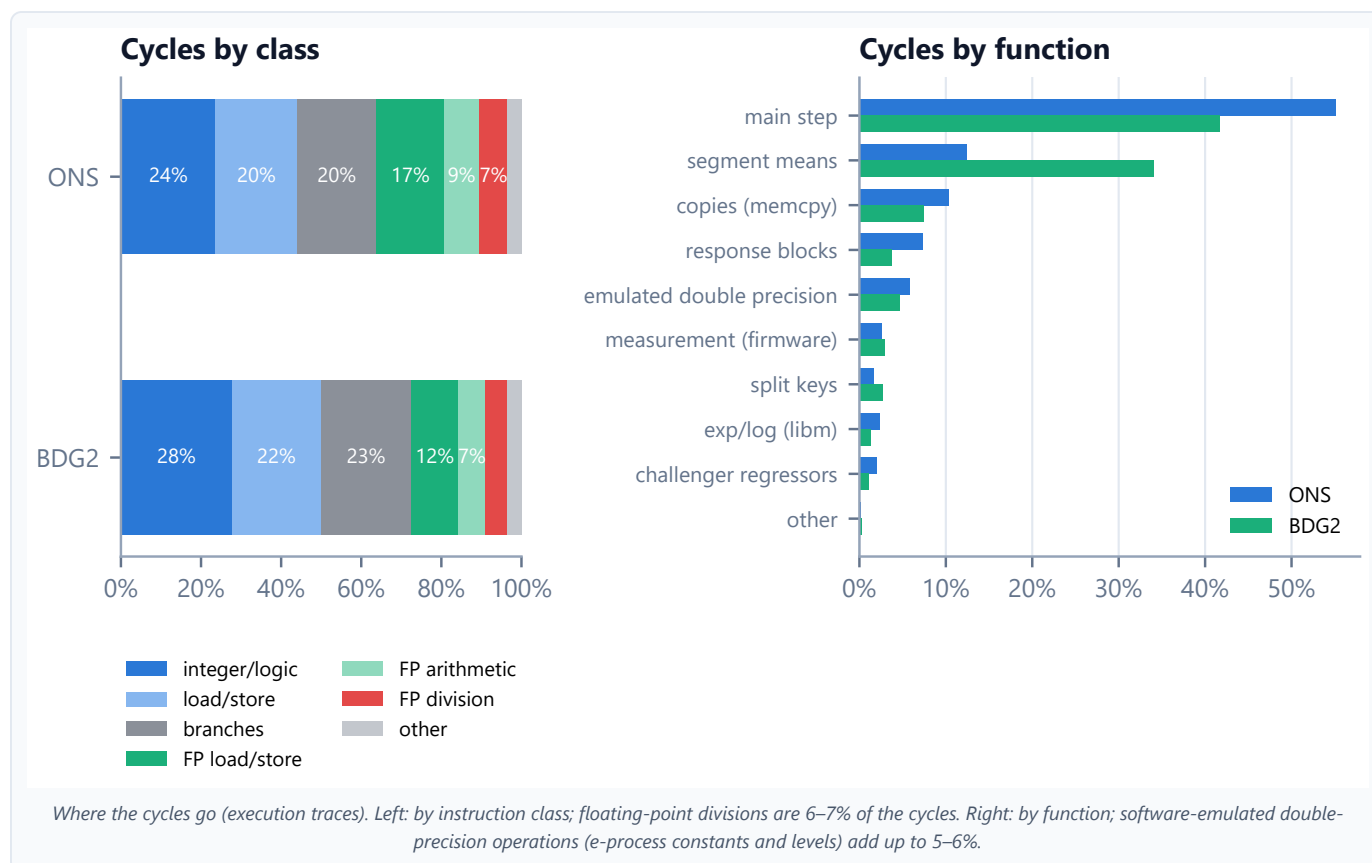
15.2 Per-step cost on the simulated device

The simulator (Renode 1.17) counts instructions executed via the Cortex-M4 DWT counter, but does not model the pipeline. Cycles were estimated from execution traces (~ 300 steps in the middle of two series) with the published Cortex-M4 and FPU timings, assuming zero-wait flash (prefetch accelerator): **CPI 1.62 and 1.57**. Times in μs below use CPI 1.6 and are estimates.



Series	Source	Inputs	Mean	Median	p99.9	Peak	Cycles (×1.6)	Time	Scaler	NMSE	Changes
ONS (dev.)	dev.	1	3,703	3,363	12,327	42,021	~5,925	~35 μs	22	0.2826	4
CAMELS (dev.)	dev.	3	4,560	4,599	14,387	48,447	~7,296	~43 μs	110	0.0229	3
BDG2 (dev.)	dev.	2	5,964	5,607	16,843	45,654	~9,542	~57 μs	33	0.0305	2
CAMELS 51795000	reserve 3	3	3,203	3,003	12,223	36,771	~5,124	~31 μs	110	0.1465	0
CAMELS 64625000	reserve 3	3	3,521	3,215	12,267	36,708	~5,634	~34 μs	110	0.3249	0
BDG2 Fox education Ollie	reserve 3	2	5,806	5,587	15,375	45,402	~9,290	~55 μs	34	0.0162	3
BDG2 Panther education Vincent	reserve 3	2	5,528	5,463	16,971	45,696	~8,844	~53 μs	33	0.0584	3





Reading: ~3.2–6.0 thousand instructions per step correspond to ~9–15 instructions per analytic operation (memory accesses, loops, indices, divisions). At ~31–57 μ s per step, a 1-second sampling would take < 0.01% of the core time; energy, estimated from an order-of-magnitude datasheet figure (~40 mA at 3.3 V), would be ~5–7 μ J per step — **not measured**. The ~73 KB state fits the STM32F407 RAM, but not smaller microcontrollers without reducing capacities (the hypothesis table dominates). There is clear optimisation room (divisions, double-precision emulation, structure copies), not explored in this version.

16. Ablations: what each part contributes

Ablations evaluated together with the reserves (same series, same mask). Earlier versions appear here only as ablations.

Variant	What changes	Result
All on, no tests	all inputs and the latent state active from the start	reserve 1, without the 6 gap series (exploratory): 0.713 vs 0.706; paired ratio 0.99 — the testing mechanism does not improve accuracy
No hierarchical splits	the response stays at octave bands + peak	ratio 0.996 [0.988; 1.001] — no difference on real data
Per-lag hypothesis ("atomic")	one hypothesis per lag of each input (~400 hypotheses)	development: 0.757 vs 0.645 (1,000 points); slower discovery
No output safeguards	open-loop memory during gaps	reserve 2: 1.107 vs 0.767; 2 catastrophic; paired ratio 0.693
Base with step 0.1	larger reference misadjustment	10/40 acceptances on structureless targets (§7)
Previous architecture (v0.51)	two experts with per-lag discovery, no stream control	reserve 3: 2.160, 4 catastrophic

Consequence for what can be claimed: LEBRE's accuracy comes mainly from a cheap online linear regression on the current inputs and a seasonal memory, protected by output safeguards. The mechanism of tested changes **is not justified by an accuracy gain**; it is justified, if at all, by the *certified and logged* structure (which inputs, with which delay, since when, with which evidence) without a cost in accuracy.

17. Position in the literature

Aspect	Related work	Classification
Structural change accepted by an e-process of predictive improvement, with stream error control	Amoukou, Mishra & Veloso (2026), in online decision trees (FWER control); Choe & Ramdas (2024); Xu & Ramdas (2024)	not original
Per-input units, hierarchical refinement, tested removal and swap, FDR control, fixed budget, on series with inputs	known parts (hierarchical testing; online group selection; e-LOND)	integration/extension
Condition of use (reference misadjustment)	Giacomini & White (2006); Clark & West (2007); LMS misadjustment (Widrow & Stearns 1985); Han & Qu (2026)	modest methodological observation, derived from known results
Information limit on smooth inputs	Stambaugh (1999); Campbell & Yogo (2006)	known; illustrative quantification
Forecasting with a budget of ~10 ² operations per step	SparseTSF, FITS, TTM, TinyCast and other small models, generally univariate and batch-trained	empirical contribution, within the tested scope
Hold, output contract, cost limits	engineering practice	standard

The literature search was limited (one search engine, English queries, full reading of few works); relevant recent works may have been missed.

18. Evolution of the architecture

LEBRE started (v0.1, the project's canonical reference) as an architecture in which structure enters and leaves the model by evidence, within a budget. v0.51 used two experts with per-lag discovery and martingale/CUSUM tests; later analyses showed that, on real series, almost all of the result came from the memory. v0.52 kept the memory and the combination, and replaced discovery by a generic engine of tested changes. The internal evolution of v0.52 was:

- per-lag hypotheses (~400) → **per whole input** (fewer hypotheses; better on real data);
- bands diluted isolated delays → **peak lag** and hierarchical splits;
- slow evidence → **one-sided c** (reading of the proof) and persistent peak statistic;
- acceptances on flat stretches → loss **scale floor**;
- acceptances on structureless targets → **step 0.05** in the reference (§7);
- cost peaks → one warm-up at a time, spread screen, half-window peak statistic;
- explosions in gaps (reserve 1) → **hold and output contract**.

19. Limits and claims we do not make

- **Domains:** hydrology (ONS, CAMELS-BR) and buildings (BDG2), with 1 to 5 inputs; the last two reserves have no ONS. Other domains were not tested.
- **Horizon:** one step ahead only, with the current inputs available.
- **Evaluation size:** 40 and 60 series give wide confidence intervals; 4 BDG2 sites share weather with development.
- **Relative position:** “lowest error in the class” refers to the comparators and protocols used here; other comparators, configurations or protocols may change the order.
- **Testing mechanism:** does not improve accuracy; false-change control is in the practical regime, not in the exact-guarantee one (§7).
- **Discovery:** fails on very smooth inputs with medium/long delays; near-identical inputs are indistinguishable.
- **Cost:** contract exceeded by ~2–4%; microcontroller only simulated, cycles estimated, energy not measured; ~73 KB state.
- **Interval:** empirical coverage near 90%, without a formal guarantee.
- **Post-failure fix:** the safeguards were designed after seeing the failure in reserve 1; they were confirmed on two later reserves, but the design was not blind.

We do not claim that LEBRE is the best forecaster, nor the most economical, nor that its mechanism is new. We only claim what was measured: under the protocols and data described, at the cost described, the numbers reported in §12–§15.

20. Parameters

Group	Constants (the same for all series)
Units	one per input (includes the target's own past) + residual latent state (poles 0.8 and 0.95); up to 4 active
Response	bands [1], [2–3], [4–7], [8–15], [16–31] + peak lag (chosen at 200 samples) + current value; splits into halves
Challengers	RLS within episode, $P_0 = 10^{-1}$; warm-up 250; 2 slots; one warm-up at a time
Evidence	loss $\min\{e^2/B^2, 1\}$, $B = 2 \cdot \max(\hat{\sigma}_S, 0.1 \cdot s_{\text{slow}})$; $c = 1 + \max(\epsilon, 0)$; $8 \lambda = 0.9 \cdot 2^{-1/c}$; $\epsilon = +0.002$ (add, swap, split), -0.002 (remove)
Multiplicity	$\alpha = 0.05$; $\gamma = 1/(2P)$ for the first $P = 2 \cdot \text{units} $, then $0.5 \cdot \gamma_k \propto 1/(k \log^2 k)$; level $\alpha \cdot \gamma \cdot (R+1) \leq 0.5$
Episodes	scan after 100 samples; limit 5,000; wait 2,000; periodic removal every 2,000; decisions every 10 steps
Screening	prewhitened AR(1), weight 0.02, every 4 steps per part; thresholds 15.09 (χ^2_5) and 6.63 (χ^2_1)
Base of S	NLMS $\mu = 0.05$; input contract $\pm 8\sigma$ with a 31-step quarantine
Memory	NLMS $\mu = 0.05$; slow mean 0.01; profile 0.1; floors 10^{-3} and 10^{-4} ; declared cycle(s); hold during gaps
Combination and output	$\lambda = 0.99$, $\eta = 0.5$, every 8 steps; envelope with rate 10^{-3} and margin $0.5 \times$
Interval	$\alpha = 0.1$, $\gamma = 0.1$, every 8 steps
Groups	causal correlation ≥ 0.95 reported with the accepted input (does not affect decisions)

21. Pseudocode and reproducibility conditions

Simplified pseudocode of one step, sufficient to re-implement the evaluated configuration. Where text and pseudocode differ, the pseudocode together with the parameter table (§20) prevails.

```

CONSTANTS # the same for any series; nothing is tuned per series
BANDS = [1], [2-3], [4-7], [8-15], [16-31]      POLES = 0.8, 0.95      L = 31
M_MAX = 4    SLOTS = 2    WARM = 250    PEAK_N = 200    N_MIN = 100    T_MAX = 5000    COOL = 2000    REM_PERIOD = 2000
DECIDE_EVERY = 10    SCREEN_EVERY = 4    SCREEN_W = 0.02    Q_UNIT = 15.09 ( $\chi^2_5$ , 99%)    Q_SPLIT = 6.63 ( $\chi^2_1$ , 99%)
 $\epsilon$ : add = swap = split = +0.002, remove = -0.002     $\alpha = 0.05$      $\mu_S = 0.05$      $\mu_M = 0.05$     CLIP_X = 8
 $\lambda_{DMA} = 0.99$      $\eta = 0.5$     EVERY = 8     $\alpha_{int} = 0.10$      $\gamma_{int} = 0.10$     SCALE_FLOOR = 0.1    OUT_MARGIN = 0.5    OUT_RHO = 1e-3

STATE # fixed size, given the number of inputs d and the period(s) s, s2
H[0..L+1][d+1] input history (includes the target's own past, standardised)
band_sum[S][d+1], seg_sum{(i,lo,hi)} running sums of the bands and of active segments
base[d+2] active(unit  $\rightarrow$  {segments, weights}) r[2] residual latent states
M: buffer, w_M, G[s], m_slow    hyps{key  $\rightarrow$  e-process}    slots[ $\leq 2$ ]

STEP(x_t, y_t or gap, quarantine):
# 1. read
 $\tilde{x} \leftarrow [x_t, \text{std\_causal}(y_{last})]$  # EMA mean/variance 1e-4, start (0, 1)
if any  $|\tilde{x}| > \text{CLIP\_X}$ : quar_until  $\leftarrow t + L$ ;  $\tilde{x} \leftarrow \text{clip}(\tilde{x}, \pm \text{CLIP\_X})$ ; shift H, band_sum, seg_sum
# 2. predict
 $\hat{y}_M \leftarrow M.\text{predict}()$  # last value + w_M  $\cdot \phi$  (reversion, difference, seasonal, profile)
 $\hat{y}_S \leftarrow \text{base}[1, \tilde{x}] + \Sigma_u w_u \cdot \text{block}(u)$  # block: means of the active segments
 $\hat{y} \leftarrow \hat{y}_M + w_S (\hat{y}_S - \hat{y}_M)$ 
if n_obs  $\geq 50$ :  $\hat{y} \leftarrow \text{clip}(\hat{y}, \text{env\_lo} - 0.5 \cdot W, \text{env\_hi} + 0.5 \cdot W)$ ,  $W = \text{env\_hi} - \text{env\_lo}$ ; if  $\hat{y}$  not finite:  $\hat{y} \leftarrow y_{last}$ 
for each experiment e:  $g_e \leftarrow \hat{y}_S + \theta_e \cdot z_e - (\text{removed contribution})$ 
return  $\hat{y}$ , [ $\hat{y} - \hat{q}$ ,  $\hat{y} + \hat{q}$ ]
# 3. observe and learn (only if y_t is observed)
learn  $\leftarrow y_t$  observed  $\wedge \neg \text{quarantine} \wedge t > \text{quar\_until}$ 
 $B \leftarrow 2 \cdot \max(\sigma_r, \text{SCALE\_FLOOR} \cdot \text{sd\_slow}(y))$ ;  $\ell(u) = \min((y_t - u)^2 / B^2, 1)$ 
if learn:
  for each e:  $k_e \leftarrow 1$ 
    if  $k_e > \text{WARM}$ : e.hyp.observe( $\ell(\hat{y}_S) - \ell(g_e) - \epsilon_e$ ) # evidence
    if  $k_e \leq \text{PEAK\_N}$ : accumulate prewhitened correlation per lag (half the lags per step)
    if  $k_e = \text{PEAK\_N}$ :  $k^* \leftarrow \text{argmax}_k c_{xy^2}/c_{xx}$ ; if  $k^* > 1$ : insert  $x_i(t-k^*)$  into the challenger
    RLS( $\theta_e, P_e$ ;  $z_e$ ,  $\text{alvo} = y_t - \hat{y}_S + \text{removed contribution}$ ) #  $P_0 = 10 \cdot I$ 
    NLMS(base, w_u;  $\text{res} = y_t - \hat{y}_S$ ,  $\mu_S$ ,  $\text{norm} = \|[1, \tilde{x}]\|^2 + \Sigma \|\text{block}(u)\|^2$ )
    screen_spread(res) # each part on its own phase, every 4 steps
   $D \leftarrow \lambda_{DMA} \cdot D + ((y_t - \hat{y}_S)^2 - (y_t - \hat{y}_M)^2) / \sigma^2$ ; every EVERY:  $w_S \leftarrow 1 / (1 + \exp(\eta \cdot D))$ 
  interval: miss  $\leftarrow |y_t - \hat{y}| > \hat{q}$ ; every EVERY:  $\hat{q} \leftarrow \max(0, \hat{q} + \gamma_{int} \cdot \sigma \cdot \Sigma(\text{miss} - \alpha_{int}))$ 
  M.update( $y_t$  if observed; otherwise the last observed value (hold), without learning)
  env  $\leftarrow$  envelope of observed values (max/min relaxing to the mean, rate OUT_RHO)
   $r \leftarrow \text{POLES} \cdot r + (1 - \text{POLES}) \cdot (y_t - \hat{y}) / \sigma$ 
  if  $t \bmod \text{DECIDE\_EVERY} = 0$ : DECIDE()

DECIDE():
  for each e: if  $\log E(e.\text{hyp}) \geq \log(1/\alpha_{hyp})$ : ACCEPT(e)
    else if  $n_e \geq T\_MAX$  or ( $n_e \geq N\_MIN$  and  $\Sigma_{\text{episode } d \leq 0}$ ): close (the evidence stays with the hypothesis)
  close experiments about units that just changed; FILL()

FILL(): while a slot is free:
  1. active unit without a removal test for  $\geq \text{REM\_PERIOD} \rightarrow$  open remove
  2. if some challenger is warming up: stop # one warm-up at a time
  3.  $u^* \leftarrow \text{argmax stat}(u)$  among inactive, free units out of cool-down;  $s^* \leftarrow \text{argmax stat}(splits)$ 
  4. pick the larger of  $\text{stat}(u^*)/Q\_UNIT$  and  $\text{stat}(s^*)/Q\_SPLIT$  (if  $\geq 1$ ):
    split  $\rightarrow split$ ; |active|  $< M\_MAX \rightarrow add$ ; else  $\rightarrow swap$  (with the unit of smallest contribution)
  # a new hypothesis n receives a fixed level:  $\alpha_n = \min(\alpha \cdot \gamma_n \cdot (R+1), 0.5)$ ,
  #  $\gamma_n = 1/(2P)$  for  $n \leq P = 2 \cdot |\text{units}|$ , then  $0.5 \cdot C / ((n-P) \cdot \log^2(n-P+1))$ ; R = changes accepted so far

e-process( $\epsilon$ ):  $c = 1 + \max(\epsilon, 0)$ ;  $\lambda_j = 0.9 \cdot 2^{-(j-1)} / c$ ,  $j = 0..7$ ;  $\psi_j = (-\log(1 - c\lambda_j) - c\lambda_j) / c^2$ 
  observe(d):  $\gamma \leftarrow \text{clip}(\text{past mean of } d, -c, 0)$ ;  $S \leftarrow S + d$ ;  $V \leftarrow (d - \gamma)^2$ 
  log E =  $\log((1/8) \cdot \Sigma_j \exp(\lambda_j \cdot S - \psi_j \cdot V))$ 

ACCEPT(e): add/swap  $\rightarrow$  active[i]  $\leftarrow (\text{BANDS} \cup \{[k^*]\}, \theta_e[-1])$ ; base[i]  $\leftarrow \theta_e[-1]$ ; remove  $\rightarrow$  delete active[u]
  split  $\rightarrow$  replace band [lo,hi] by its two halves with the challenger weights; R  $\leftarrow R + 1$ 

```

Reproducibility conditions

- Inputs:** standardise each input causally before the model: $z = (x - m) / \sqrt{v + 10^{-6}}$, with exponential m and v at rate 10^{-4} from (0, 1), updated *after* standardising the current value; missing values become 0 after standardisation.

- **Target:** passed in original units; missing values are passed as a gap (no learning); an external quarantine flag suspends learning.
- **Metadata:** seasonal period s (24 for hourly data; none if there is no cycle) and, optionally, a second cycle s_2 (168 for buildings).
- **Determinism:** the algorithm uses no random numbers; given the same input sequence, the output is deterministic. In single precision, decisions near the threshold may shift by a few steps ($\approx 10\%$ of the series), without material effect on the error.
- **Evaluation:** split of series by a fixed-seed permutation before any modelling; test region = last 70%; metric relative to the online NLinear calibrated on the same series; geometric mean and stratified bootstrap (fixed seed).
- **Environment used:** Python 3.11, NumPy, pandas 2.2.3 (later versions changed read-only array behaviour); C99 with Arm GCC 15.2 (–O2, single-precision FPU, no FMA contraction); Renode 1.17.

22. References

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